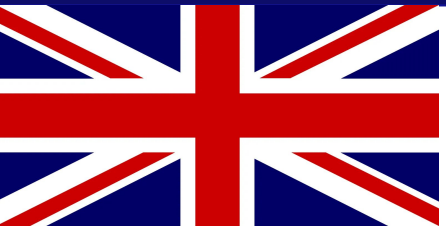
Data Logging and Classification

Where engineering meets AI



UNIVERSITY OF SUSSEX

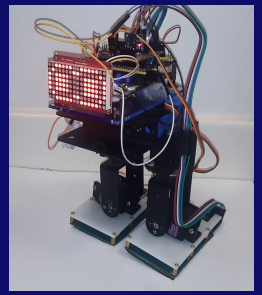
Introduction

Dexter Shepherd

Robotics researcher at the University of Sussex, Brighton

Lead for international AI mentoring program

德克斯特·谢泼德
布莱顿 Brighton United Kingdom University of Sussex
机器人研究员
国际人工智能导师项目负责人



University of Sussex

Interdisciplinary teaching and research, encouraging collaboration across different fields.
Strong global reputation, consistently ranked among leading universities worldwide.
Beautiful green campus located within the South Downs National Park.
Close to the vibrant seaside city of Brighton, known for its culture, music, and creativity.
Diverse international community with students and staff from over 100 countries.
Active student life with hundreds of societies, sports clubs, and events.

跨学科的教学与研究，鼓励不同领域之间的合作。
享有强大的国际声誉，持续位列全球领先大学之列。
美丽的绿色校园，坐落于南唐斯国家公园内。
靠近充满活力的海滨城市布莱顿，以其文化、音乐和创造力而闻名。
多元化的国际社区，拥有来自100多个国家的学生和教职员工。
丰富的学生生活，拥有数百个社团、体育俱乐部和各类活动。

Engineering and Informatics at Sussex

Great environment full of social events and hackathons

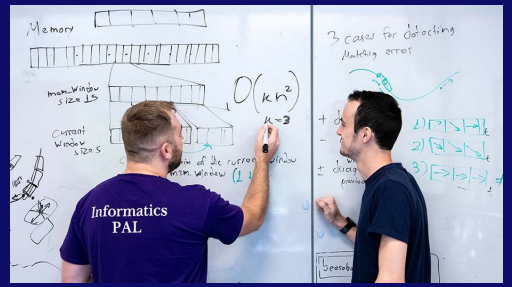
Research led modules, benefiting directly off expert cutting edge work

Mentor support schemes to help you if you fall behind

社交活动和黑客松丰富的优秀环境

以研究为主导的课程模块，能够直接受益于专家的前沿研究成果

导师支持计划，在你进度落后时为你提供帮助



Today’s task

Building fitness trackers

How do smart watches work?

Accelerometers monitor acceleration in various directions

Smart watches track this and use machine learning models to interpret this information

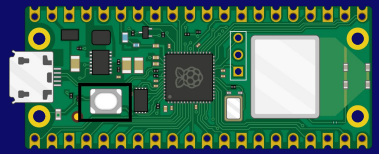
From these movements, models can classify what sort of exercise activity is being done.

加速度计可以监测不同方向上的加速度。
智能手表会跟踪这些数据，并使用机器学习模型来解读这些信息。
通过这些运动数据，模型可以分类判断正在进行的运动类型。

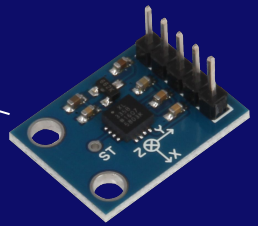
Key Components for fitness

Screen / 屏幕 或 显示屏

Controller / 控制器



Accelerometer / 加速度计

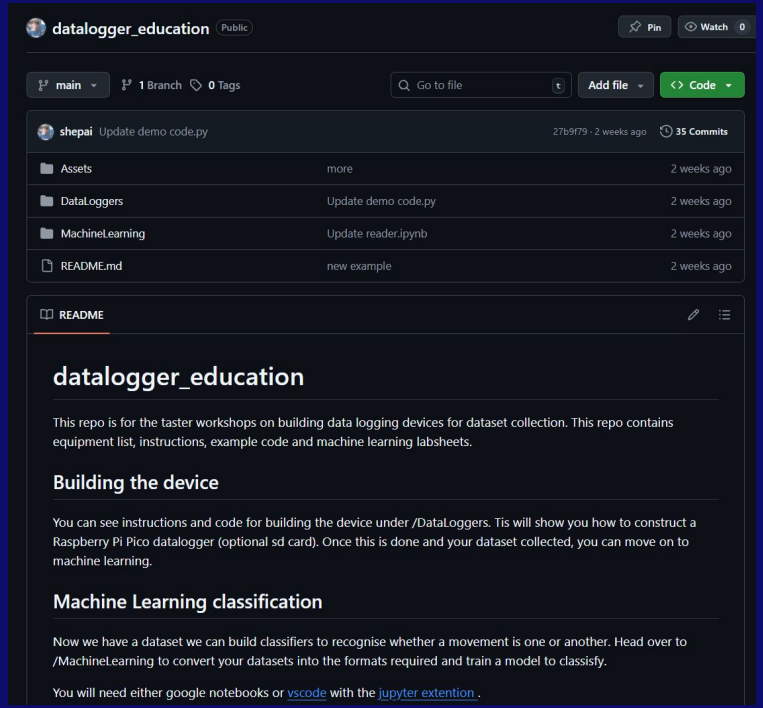


Todays task

https://github.com/shepai/datalogger_education

Go to the website and follow the instructions for building the raspberry pi device

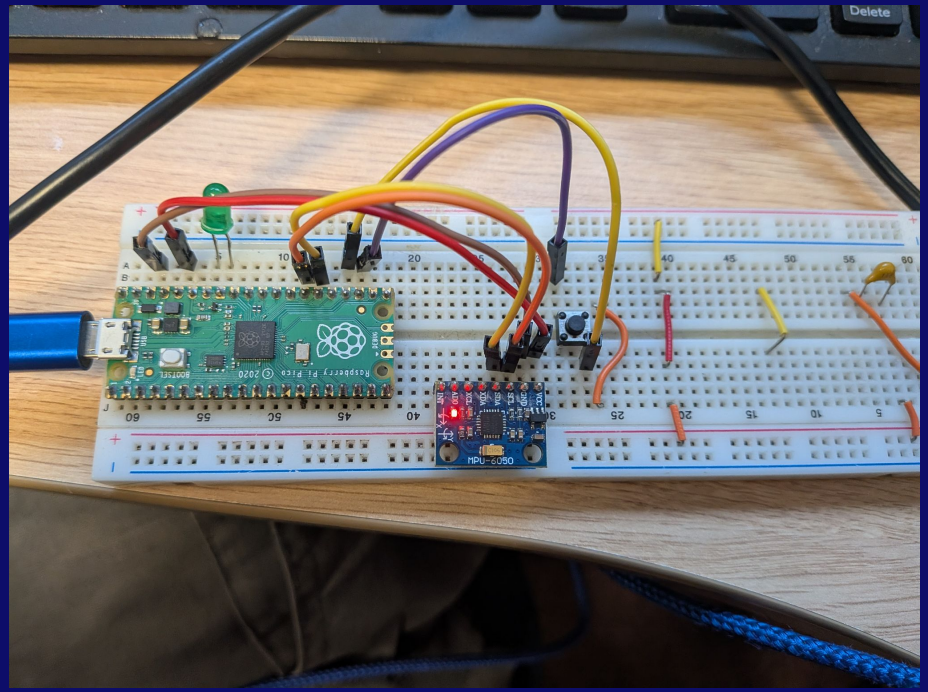
访问该网站，并按照说明构建 Raspberry Pi 设备。



End result

You will need to download Thonny IDE

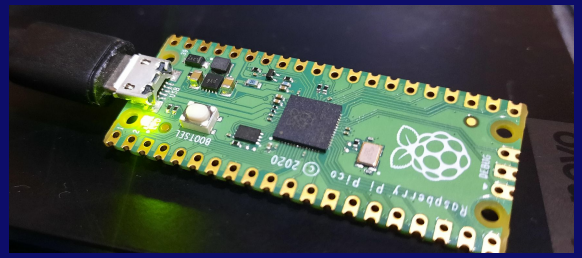
你需要下载 Thonny 集成开发环境 (IDE)

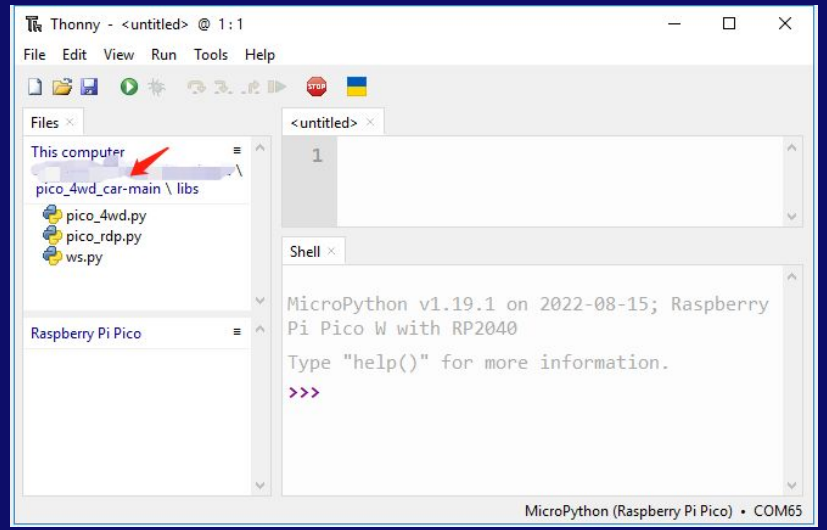


Uploading code

Plug it in via USB - 通过 USB 将它连接上。

Open Thonny IDE - 打开 Thonny 集成开发环境 (IDE)。

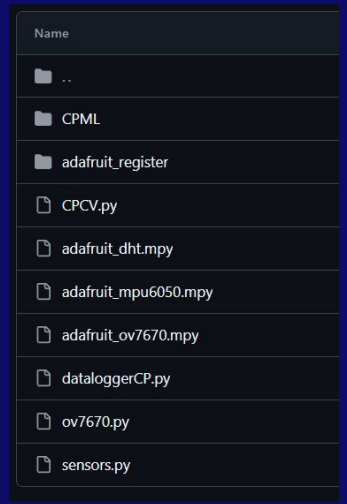




Uploading code

In the GitHub there is a folder called CircuitPython dependencies. Copy and paste the content into the /lib directory

在 GitHub 上有一个名为 “CircuitPython dependencies” 的文件夹。将其中的内容复制并粘贴到 /lib 目录中。



Uploading code

Copy the contents over from demo code and paste it into code.py on the device

将 demo 代码中的内容复制过来，并粘贴到设备上的 code.py 文件中。

datalogger_education / DataLoggers / demo code.py

shepai Update demo code.py

Code Blame 45 lines (43 loc) · 1.16 KB

```python
#import the data logging lib
import busio
import digitalio
import board
from dataloggerCP import datalogger
import adafruit_mpu6050
import time

#set up the pins
spi = busio.SPI(board.GP2, board.GP3, board.GP4)
cs = board.GP1
i2c = busio.I2C(board.GP21,board.GP20) #might need to flip it
btn=digitalio.DigitalInOut(board.GP18)
btn.direction=digitalio.Direction.INPUT
btn.pull=digitalio.Pull.UP
LED=digitalio.DigitalInOut(board.GP28)
LED.direction=digitalio.Direction.OUTPUT
#set up the sensor
mpu = adafruit_mpu6050.MPU6050(i2c)
logger=datalogger(spi,cs)
#read line function from the sensor
def read():
    xa,ya,za=mpu.acceleration
    xg,yg,zg=mpu.gyro
    return str(xa)+","+str(ya)+","+str(za)+","+str(xg)+","+str(yg)+","+str(zg)+"\n"
```

Gathering datasets

Think of at least two actions (for example running and walking)

Press the button on your circuit to begin recording, and take note of what you did while recording data. Press the button again to stop recording.

Record a few actions and then back them up and label them on your computer

想出至少两个动作（例如跑步和走路）。

按下电路板上的按钮开始录制，并在录制数据时记下你做了什么动作。再次按下按钮停止录制。

录制几个动作，然后将数据备份到你的电脑上并进行标注。

Preview Code Blame 12355 lines (12355 loc) · 421 KB

Search this file

1	-4199	324	-7788	1098	970	-811
2	-4227	472	-7580	911	1055	-854
3	-4183	400	-7640	632	1025	-862
4	-4159	292	-7632	592	1120	-799
5	-4203	300	-7356	671	1109	-772
6	-4227	264	-7208	737	1025	-743
7	-4215	216	-7188	826	857	-708
8	-4099	288	-7084	839	824	-614
9	-4059	132	-7084	791	684	-578
10	-3939	76	-7056	774	650	-475
11	-3927	76	-7020	721	654	-482
12	-3983	96	-6980	733	614	-506
13	-3995	28	-6844	727	527	-604

Machine learning

Once we have gathered the data sets we can make classifiers

I suggest downloading vscode for this task, and install the jupyter extension

We will be working from something called notebooks

一旦我们收集好了数据集，就可以构建分类器。

我建议为此下载 VS Code，并安装 Jupyter 扩展。

我们将使用一种叫做 notebooks（笔记本）的方式来进行工作。

What is a classifier?

an algorithm that automatically categorizes data into one or more predefined classes

一种能够将数据自动分类到一个或多个预定义类别中的算法。

What can you classify

Anything that is made up of data!

Sensor readings: analogue float values

Pictures: arrays of pixel numbers

Videos: arrays of pictures

Text: represented as vector space for large language models

任何由数据构成的东西！

传感器读数：模拟浮点数值
图片：像素数值数组
视频：由多张图片组成的数组
文本：在大型语言模型中表示为向量空间中的向量

How we will be using it

We will be using simple classifiers to classify our datasets. Work through the code on the Github.

How does your dataset do? Can you find ways to improve it?

Things to look into:

PCA and data augmentation

Different models

我们将使用简单的分类器来对我们的数据集进行分类。请按照 Github 上的代码一步步运行。
你的数据集表现如何？你能找到改进的方法吗？

可以研究的内容：

- PCA 和数据增强
- 不同的模型

datalogger_education / MachineLearning /